COMMODITY ANALYST

Hormuz Fertilizer Disruption Eases, But Risks Remain

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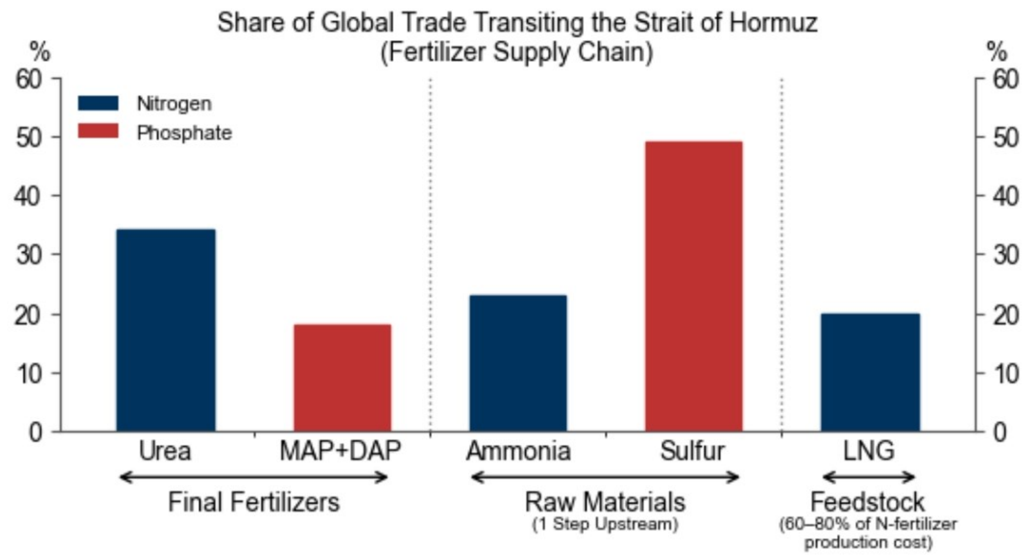
- While the immediate fertilizer disruption has eased following the reopening of the Strait of Hormuz (through which one-third of global fertilizer trade flows), risks remain: any renewed disruption to Strait flows could be consequential if it coincided with the Q3 procurement season for major nitrogen fertilizer importers ahead of 2H planting, while phosphate affordability continues to pressure Brazilian soybean farmers.
  - **Nitrogen fertilizers**—critical for grains such as corn, wheat, and rice—have an immediate and direct impact on yields and must be applied each season, making timing critical and supply disruptions particularly damaging. The Q2 disruption to flows through the Strait of Hormuz was therefore relatively contained, as it occurred during a seasonal lull—after Northern Hemisphere farmers had largely secured supply and before the next procurement cycle began. Prices nonetheless rose by ~50% at the onset of the closure of the Strait and have since retraced in June as flows resumed amidst seasonally soft demand. However, this same inflexibility makes nitrogen the key near-term risk if disruptions were to re-emerge in Q3 when major importers procure for planting, including Brazil corn, India rice and sugarcane, and EU winter wheat.
  - **Phosphate fertilizers**, by contrast, can be drawn down from soil reserves in the short term, limiting the immediate impact on yields, but sustained under-application depletes these reserves and gradually weakens soil quality, crop resilience, and productivity. This is particularly relevant in Brazil soybeans (accounting for 62% of global exports), where import dependence on phosphate fertilizers is high (~80%). June-July is a critical procurement window for Brazil soybeans ahead of September planting, yet soybean growers had secured only ~68% of expected fertilizer demand by mid-June (vs. ~75% historically). Although Strait of Hormuz fertilizer flows have resumed, prices have not declined significantly. With affordability already constrained since the 2022 Russia-Ukraine disruption, the latest shock risks keeping Brazilian soybean farmers under pressure, potentially prolonging potential under-application and raising longer-term risks to Brazilian soybean production.

Hormuz Fertilizer Disruption Eases, But Risks Remain

Fertilizers came into focus following the disruption at the Strait of Hormuz, through which roughly one-third of global fertilizer trade flows (Exhibit 1). The disruption tightened supply through two channels: directly, via reduced exports of finished nitrogen and phosphate fertilizers, and indirectly, through constrained access to key upstream inputs — including sulfur for phosphate and ammonia and natural gas for nitrogen— limiting substitution across regions.

Exhibit 1: The Strait of Hormuz Closure Tightened Fertilizer Supply Both Directly—Via Reduced Exports of Finished Nitrogen and Phosphate—and Indirectly by Constraining Access to Its Upstream Inputs

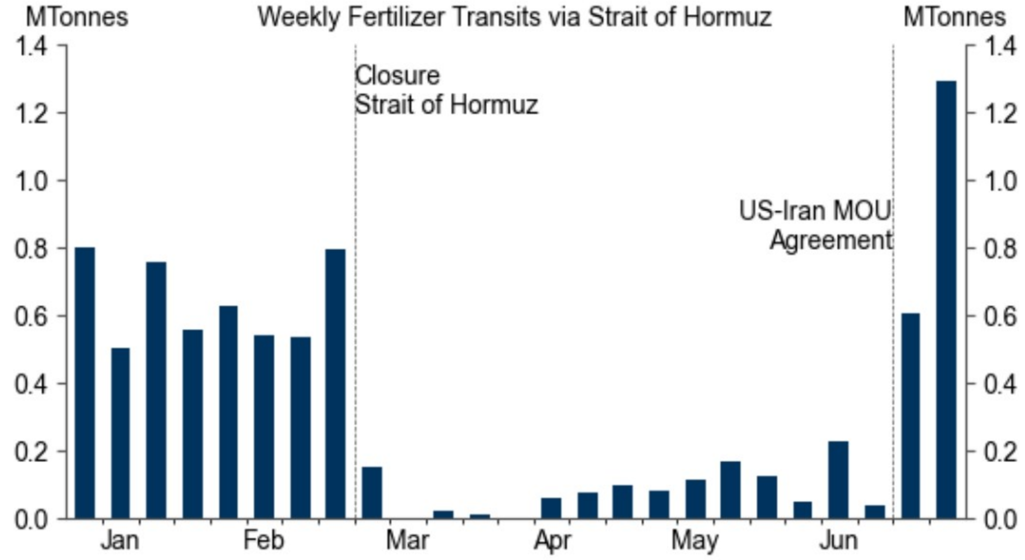


Trade shares for 2024. MAP stands for monoammonium phosphate. DAP stands for diammonium phosphate.

Source: IFA, Goldman Sachs Global Investment Research

Although flows have so far recovered since mid-June (Exhibit 2), the situation remains fragile, leaving agriculture exposed to renewed fertilizer disruptions.

Exhibit 2: Fertilizer Flows Out of the Strait of Hormuz Have Picked Up Since Its Reopening



Source: Kpler, Goldman Sachs Global Investment Research

The risk differs by fertilizer type. **Nitrogen fertilizers**—critical for grains such as corn, wheat, and rice—have an immediate and direct impact on yields and must be applied each season, making timing critical and supply disruptions particularly damaging. The Q2 disruption to flows through the Strait of Hormuz was therefore relatively contained, as it occurred during a seasonal lull—after Northern Hemisphere farmers had largely secured supply in Q1 and before the next procurement cycle begins in H2.

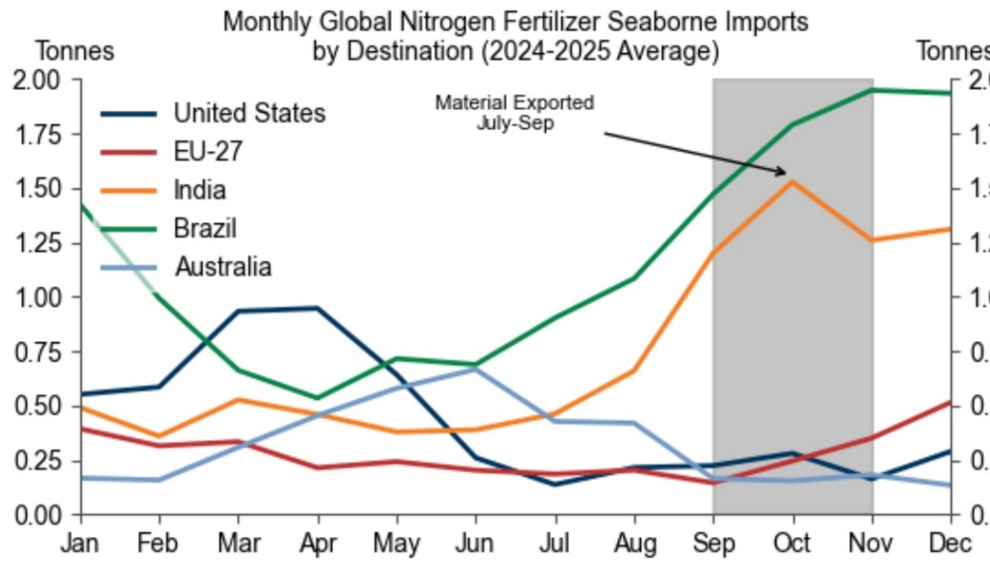
Prices for urea (the main nitrogen fertilizer) nonetheless rose by roughly 50% at the onset of the closure of the Strait and have since retraced in June as flows resumed amidst still seasonally soft demand (Exhibit 3, Exhibit 4).

Exhibit 3: Urea Prices Rose ~50% at the Onset of the Strait's Closure to Its Peak But Have Since Retraced

Source: Bloomberg, Goldman Sachs Global Investment Research

However, this same inflexibility makes nitrogen the key near-term risk if disruptions re-emerge, particularly in Q3 when major importers procure for planting, including Brazil corn, India rice and sugarcane, and EU winter wheat (Exhibit 4).

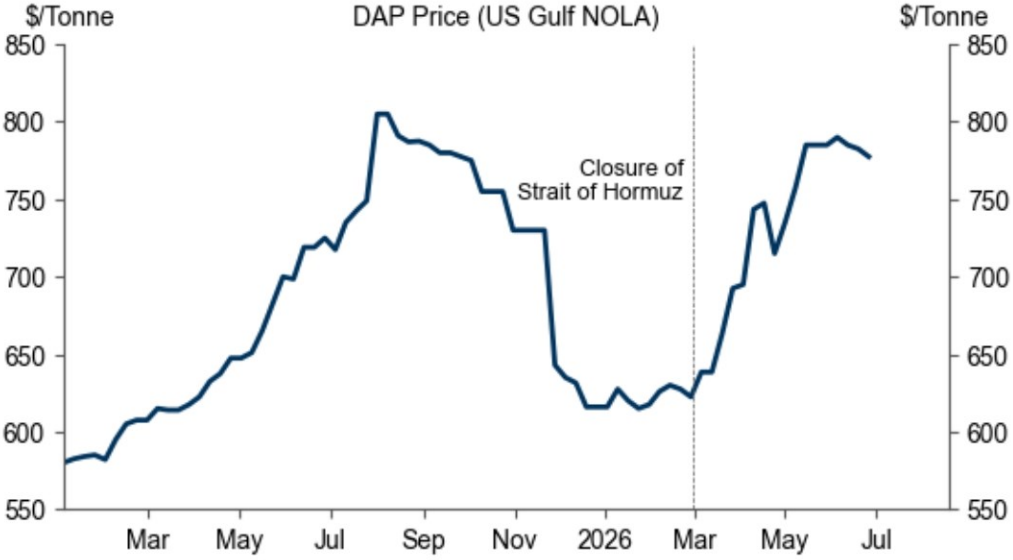
Exhibit 4: Renewed Nitrogen Disruptions Are a Key Risk in Q3 When Major Nitrogen Importers Procure for Planting



Source: Kpler, Goldman Sachs Global Investment Research

Phosphate, by contrast, can be drawn down from soil reserves in the short term, limiting the immediate impact on yields, but sustained under-application depletes these reserves and gradually weakens soil quality, crop resilience, and productivity. This is particularly relevant in Brazil soybeans (accounting for 62% of global exports), where import dependence on phosphate fertilizers is high (~80%), June-July is a critical procurement window ahead of September planting, yet soybean growers had secured only ~68% of expected fertilizer demand by mid-June (vs. ~75% historically). Although Strait of Hormuz fertilizer flows have resumed, prices have not declined significantly (Exhibit 5).

Exhibit 5: Phosphate Prices Remain Elevated Despite the Reopening of the Strait of Hormuz

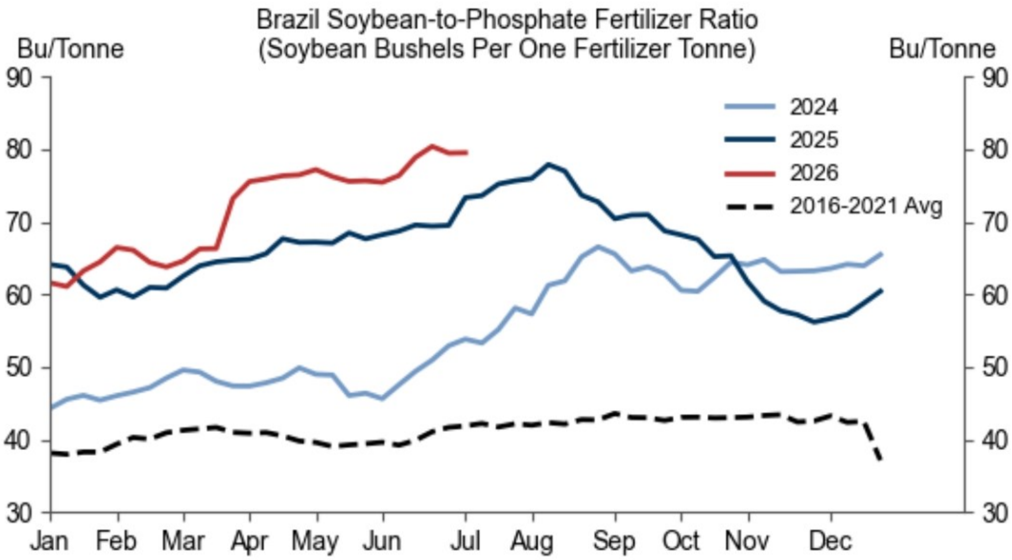


DAP stands for diammonium phosphate.

Source: Bloomberg, Goldman Sachs Global Investment Research

With affordability already constrained since the 2022 Russia-Ukraine disruption, the latest shock risks keeping Brazilian soybean farmers under pressure (Exhibit 6), potentially prolonging potential under-application and raising longer-term risks to Brazilian soybean production.

Exhibit 6: Phosphate Fertilizers Have Become Increasingly Less Affordable Since 2022



Source: Bloomberg, Goldman Sachs Global Investment Research

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Disclosure Appendix

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